SKILL - 9 Global Exception Handling using @ControllerAdvice

Code:

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Name: S. Gowtham

Skill Experiment Number:09

Section Number:75

Skill9Application.java

```
package com.example.demo;


import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;


@SpringBootApplication
public class Skill9Application {


	public static void main(String[] args) {
		SpringApplication.run(Skill9Application.class, args);
	}


}
```

StudentController.java

```
package com.example.demo.controller;


import com.example.demo.model.Student;
```

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```
import com.example.demo.exception.StudentNotFoundException;
import com.example.demo.exception.InvalidInputException;

import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/student")
public class StudentController {

  @GetMapping("/{id}")
  public Student getStudent(@PathVariable int id) {

    if(id <= 0) {
      throw new InvalidInputException("Student ID must be positive");
    }

    if(id != 1) {
      throw new StudentNotFoundException("Student not found with ID: " + id);
    }

    return new Student(1, "Saranya", "Computer Science");
  }
}
```

ErrorResponse.java

```
package com.example.demo.controller;
```

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```
import com.example.demo.model.Student;
import com.example.demo.exception.StudentNotFoundException;
import com.example.demo.exception.InvalidInputException;

import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/student")
public class StudentController {

  @GetMapping("/{id}")
  public Student getStudent(@PathVariable int id) {

    if(id <= 0) {
      throw new InvalidInputException("Student ID must be positive");
    }

    if(id != 1) {
      throw new StudentNotFoundException("Student not found with ID: " + id);
    }

    return new Student(1, "Saranya", "Computer Science");
  }
}
```

GlobalExceptionHandler.java

```
package com.example.demo.exception;
```

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```
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ControllerAdvice;
import org.springframework.web.bind.annotation.ExceptionHandler;

import java.time.LocalDateTime;

@ControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(StudentNotFoundException.class)
    public ResponseEntity<ErrorResponse>
handleStudentNotFound(StudentNotFoundException ex) {

        ErrorResponse error = new ErrorResponse(
                LocalDateTime.now(),
                ex.getMessage(),
                HttpStatus.NOT_FOUND.value()
        );

        return new ResponseEntity<>(error, HttpStatus.NOT_FOUND);
    }

    @ExceptionHandler(InvalidInputException.class)
    public ResponseEntity<ErrorResponse> handleInvalidInput(InvalidInputException ex) {
```

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```
        ErrorResponse error = new ErrorResponse(
            LocalDateTime.now(),
            ex.getMessage(),
            HttpStatus.BAD_REQUEST.value()
        );

        return new ResponseEntity<>(error, HttpStatus.BAD_REQUEST);
    }
}
```

InvalidInputException.java

```
package com.example.demo.exception;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ControllerAdvice;
import org.springframework.web.bind.annotation.ExceptionHandler;

import java.time.LocalDateTime;

@ControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(StudentNotFoundException.class)
    public ResponseEntity<ErrorResponse>
handleStudentNotFound(StudentNotFoundException ex) {
```

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```
        ErrorResponse error = new ErrorResponse(
            LocalDateTime.now(),
            ex.getMessage(),
            HttpStatus.NOT_FOUND.value()
        );

        return new ResponseEntity<>(error, HttpStatus.NOT_FOUND);
    }

    @ExceptionHandler(InvalidInputException.class)
    public ResponseEntity<ErrorResponse> handleInvalidInput(InvalidInputException ex) {

        ErrorResponse error = new ErrorResponse(
            LocalDateTime.now(),
            ex.getMessage(),
            HttpStatus.BAD_REQUEST.value()
        );

        return new ResponseEntity<>(error, HttpStatus.BAD_REQUEST);
    }
}
```

StudentNotFoundException.java

```
package com.example.demo.exception;
```

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```
public class StudentNotFoundException extends RuntimeException {

  public StudentNotFoundException(String message) {
    super(message);
  }
}
```

Student.java

```
package com.example.demo.model;

public class Student {

  private int id;
  private String name;
  private String department;

  public Student(int id, String name, String department) {
    this.id = id;
    this.name = name;
    this.department = department;
  }

  public int getId() {
    return id;
  }

  public String getName() {
```

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```
        return name;
    }

    public String getDepartment() {
        return department;
    }
}
```

Output:

```
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 :: Spring Boot ::                (v4.0.3)

2026-03-09T17:14:46.885+05:30  INFO 13972 --- [StudentExceptionApp] [           main] c.e.s.StudentExceptionAppApplication     : Starting St
2026-03-09T17:14:46.912+05:30  INFO 13972 --- [StudentExceptionApp] [           main] c.e.s.StudentExceptionAppApplication     : No active p
2026-03-09T17:14:50.137+05:30  INFO 13972 --- [StudentExceptionApp] [           main] o.s.boot.tomcat.TomcatWebServer          : Tomcat init
2026-03-09T17:14:50.165+05:30  INFO 13972 --- [StudentExceptionApp] [           main] o.apache.catalina.core.StandardService   : Starting se
2026-03-09T17:14:50.165+05:30  INFO 13972 --- [StudentExceptionApp] [           main] o.apache.catalina.core.StandardEngine    : Starting Se
2026-03-09T17:14:50.262+05:30  INFO 13972 --- [StudentExceptionApp] [           main] b.w.c.s.WebApplicationContextInitializer : Root WebApp
2026-03-09T17:14:51.029+05:30  WARN 13972 --- [StudentExceptionApp] [           main] ConfigServletWebServerApplicationContext : Exception e
2026-03-09T17:14:51.060+05:30  INFO 13972 --- [StudentExceptionApp] [           main] .s.b.a.l.ConditionEvaluationReportLogger :

Error starting ApplicationContext. To display the condition evaluation report re-run your application with 'debug' enabled.
2026-03-09T17:14:51.084+05:30 ERROR 13972 --- [StudentExceptionApp] [           main] o.s.b.d.LoggingFailureAnalysisReporter   :
```

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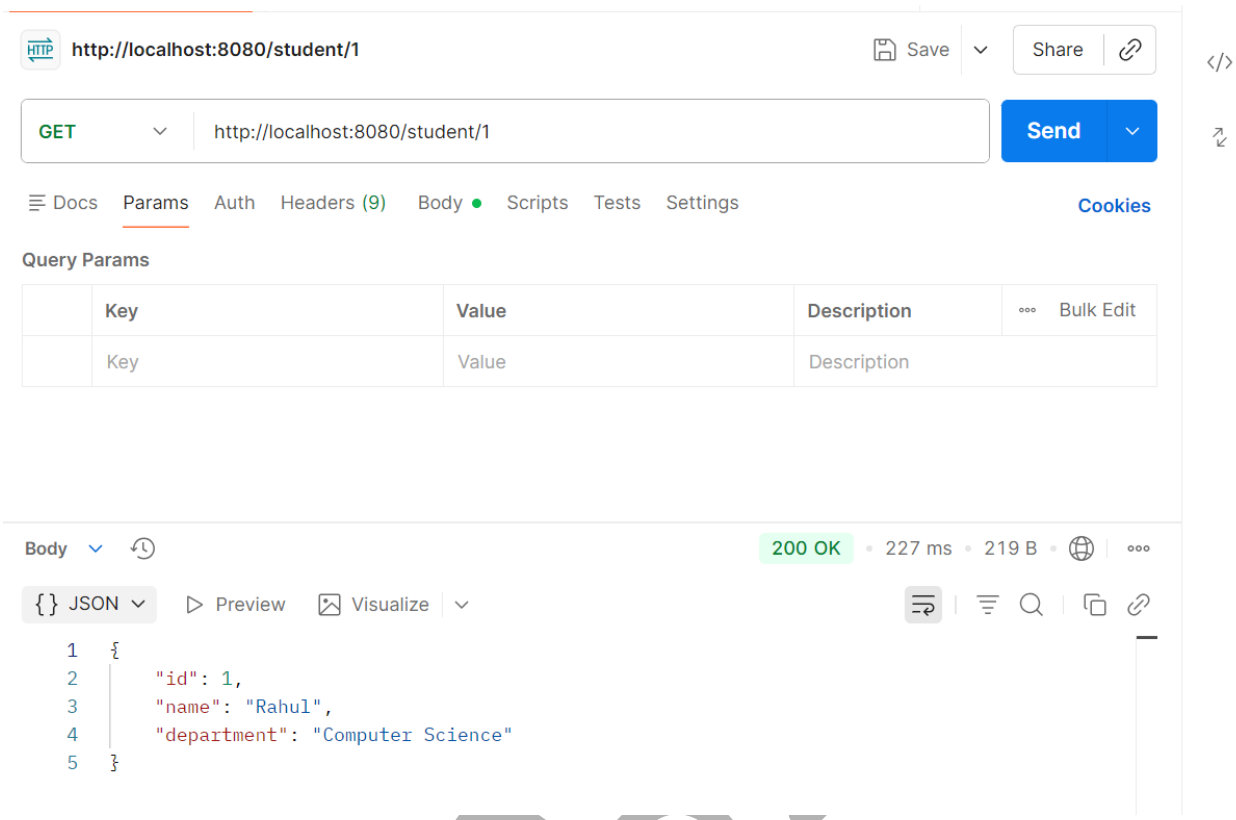

https://github.com/Gowtham-406/S-75_1041_SkillLabDocuments.git

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